Rethinking Human Cognition: A Comparative Framework Bridging Biological Roots and Social Complexity

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Abstract

Research on human cognition, behavior, and society typically studies human phenomena without first clarifying the roots from which they arise. As a result, many accounts rely on narrative interpretations and loosely defined constructs, treating the capacities behind these phenomena as culturally constructed or as emergent products of social complexity. This has contributed to unstable findings and persistent difficulties in replication. Yet decades of research in primatology and comparative cognition reveal that many of these capacities also appear in our closest living relatives, chimpanzees, suggesting that they are not human inventions but structured extensions of shared cognitive functions. Here, we propose a comparative framework that bridges biological roots and social complexity by aligning core domains of human cognition—culture and learning, cooperation and joint action, social and goal inference, power and politics, morality and fairness, and shared biological substrates—with functionally comparable capacities observed across species. This perspective reframes human cognition not as a set of isolated higher-order faculties, but as a structured continuum built from shared biological and cognitive foundations. It offers an empirical foundation for studying human cognition, moving beyond loosely defined constructs and narrative accounts toward a comparative, mechanism-based science.

Keywords: Human Cognition, Cross-Species, Vague Construct, Replication Crisis

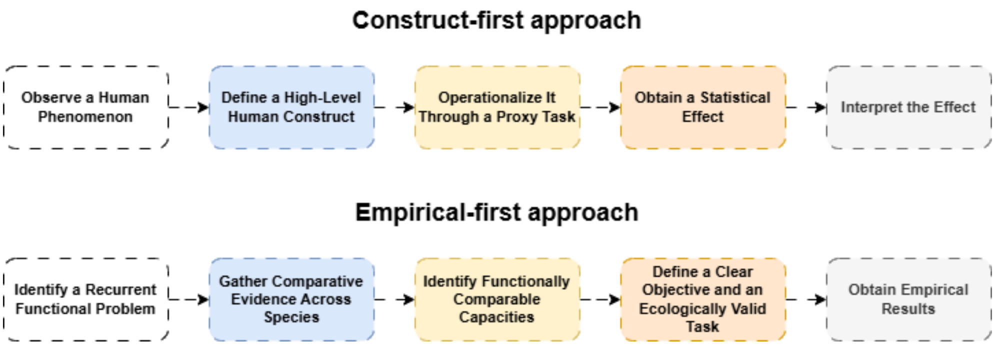

Fig. 1 Construct-first research operationalizes predefined high-level concepts through proxy tasks, leaving the resulting effects dependent on interpretation. The empirical-first approach uses comparative evidence to define capacities precisely and design ecologically grounded tasks, producing robust results.

1 Introduction

Research on human cognition, behavior, and society often begins with high-level phenomena and explains them through narrative interpretations and loosely defined constructs—such as symbolism, morality, culture, discourse, or social meaning—without identifying the more basic capacities from which they arise. Often treated as uniquely human, culturally constructed, or emergent from social complexity, these constructs may not correspond to any stable underlying capacity when operationalized. The resulting effects are therefore often small, unstable across tasks and contexts, or absent altogether, contributing to the severe and still unresolved replication crisis in research on humans [1, 2].

Yet decades of empirical work in primatology and comparative cognition have challenged this human-only view. Chimpanzees, our closest living relatives, demonstrate a wide range of cognitive behaviors long assumed to be uniquely human: they manufacture and use tools [3–5], coordinate in cooperative hunts [6–8], engage in tactical deception [9–11], express fairness-related reactions [12, 13], and navigate dynamic social hierarchies with strategic coalition-building [14–16].

Despite this growing body of evidence, primate studies are rarely organized into a systematic account. Instead, these findings often remain fragmented and are treated as remarkable observations of nonhuman animal behavior. As a result, much of the humanities and social sciences continues to rely on vague interpretive frameworks, explaining phenomena such as morality, politics, and culture without recognizing that the bottom-up logic underlying these phenomena can already be traced in primate behavior.

To address this gap, we propose a comparative framework that bridges biological roots and social complexity by aligning core domains of human cognition—culture and learning, cooperation and joint action, social and goal inference, power and politics, morality and fairness, and shared biological substrates—with functionally comparable capacities observed in chimpanzees. Rather than viewing these capacities as disconnected higher faculties, we frame them as structured elaborations of shared biological and cognitive foundations.

Our contribution.

We identify one important source of the replication crisis: many studies operationalize loosely defined constructs that do not correspond to stable underlying capacities. We therefore develop a six-domain comparative framework grounded in chimpanzee cognition, proposing that the study of human cognition should begin with functionally comparable capacities observed across species, rather than from a human-only perspective that interprets behavior through concepts defined in advance.

2 The Interpretive Inheritance of Human Science

The study of human behavior grew out of philosophy, where humans first organized their ideas about what humans were supposed to be. The apparent depth of this vocabulary often came not from empirical precision, but from conceptual openness, unstable boundaries, and the complexity produced through interpretation. Although many fields have since adopted more empirical methods, they often still lack a clear account of where human behavior comes from. As a result, empirical research often continues to begin from vague concepts inherited from this tradition, such as morality, culture, discourse, and institutions, before the origins of these behaviors have been clarified.

3 A Cross-Species Foundation for Human Science

We therefore introduce a cross-species functional taxonomy that rebuilds human cognition on an empirical basis rather than from inherited, loosely defined concepts. The taxonomy aligns key domains of human cognition with functionally similar capacities observed in chimpanzees and organizes existing evidence by underlying function rather than by species, task, or isolated findings. It provides a systematic account across six domains:

1. **Culture and Learning**: How individuals acquire, retain, and transmit knowledge through observation, interaction, and group-specific traditions — enabling behaviors to persist across time and partners.
2. **Cooperation and Joint Action**: How individuals coordinate their actions with others — working toward shared goals, dividing roles, or synchronizing behavior in collective tasks.
3. **Social and Goal Inference**: How individuals interpret others' actions, goals, and social ties — adjusting behavior according to social situations and contexts.
4. **Power and Politics**: How individuals navigate dominance, alliances, and group tensions — shaping access to resources, influence, and long-term social position.
5. **Morality and Fairness**: How individuals respond to unequal treatment, unsolicited harm, or social disruption — including fairness, helping, and repairing relationships.

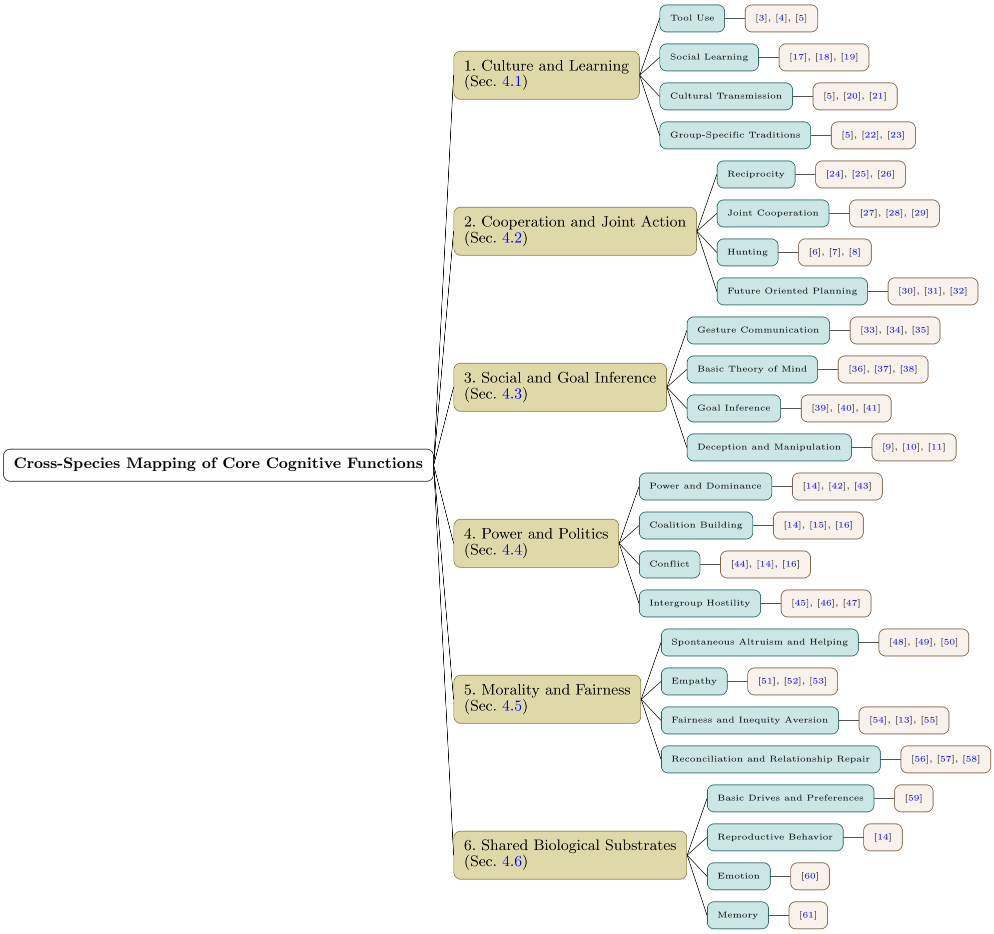

Fig. 2 Cross-species cognitive taxonomy aligning six core domains of human cognition with observed chimpanzee behaviors. Each branch includes representative subfunctions and supporting studies, illustrating the structured continuity between human and primate cognition.

6. **Shared Biological Substrates**: Common biological substrates—such as emotion, memory, and reproductive behavior—that are broadly observed across many species, not only in humans or great apes.

This taxonomy offers a functional map of primate cognition—bridging fragmented findings and reframing high-level human functions as structured extensions of biologically grounded capacities.

4 Overview of the Six Domains

In this section, we first identify the major vague constructs and conceptual problems in the humanities, and then present empirical evidence from primates to show how these concepts can be grounded in observable behavior rather than formulated through abstract interpretation.

4.1 Culture and Learning

Traditional view.

Culture has long lacked a settled definition, having variously been framed through meanings, symbols, norms, values, and shared practices [62–65]. Because culture has been studied primarily through the human case, its study has remained largely interpretive, with no clear starting point for identifying the functional components from which it arises.

Cross-species view.

Chimpanzees exhibit several constituent features of culture and learning:

- **Tool Use.** Chimpanzees use sticks, leaves, and stones to extract food, collect water, crack nuts, and perform other tasks [3–5].
- **Social Learning.** They acquire tool-use and foraging skills by observing others, and can flexibly imitate actions or reproduce their outcomes [17–19].
- **Cultural Transmission.** Chimpanzees adopt and transmit newly introduced techniques and arbitrary behaviors, allowing them to spread through groups and persist over time [20, 21].
- **Group-Specific Traditions.** Different communities maintain stable differences in tool use, foraging, and social practices, including among neighboring populations [5, 22, 23].

4.2 Cooperation and Joint Action

Traditional view.

Human cooperation has been explained through moral sentiments, social norms, reciprocal incentives, and institutions [66–69]. Because these accounts typically begin from the human case alone, their concepts often remain interpretive, leaving unclear what concrete processes they describe or at which level of explanation they operate.

Cross-species view.

Chimpanzees exhibit several constituent features of cooperation and joint action:

- **Reciprocity.** Chimpanzees exchange grooming, food, and coalition support over time, with patterns shaped by previous interactions and stable social relationships [24–26].
- **Joint Cooperation.** They recruit effective partners and coordinate their actions according to partner competence and social tolerance [27, 29, 70].
- **Hunting.** During group hunts, chimpanzees coordinate complementary roles and share meat in ways associated with collective success and alliance relationships [6–8].
- **Future-Oriented Planning.** Chimpanzees and other great apes delay gratification and retain tools for future use in the absence of immediate rewards or task cues [30–32].

4.3 Social and Goal Inference

Traditional view.

Social and goal inference have often been treated as uniquely human because they appear to involve understanding hidden intentions, meanings, and mental states [71, 72]. Since these states are not directly observable, however, such accounts often overstate what humans can actually know, turning inference from limited evidence into an apparent capacity for mind reading.

Cross-species view.

Chimpanzees exhibit several constituent features of social and goal inference:

- **Gesture Communication.** Chimpanzees use a flexible repertoire of gestures, adjusting their signals according to social context and recipient responses [33–35].
- **Social Inference.** They track what others can see and have previously encountered, using this information to anticipate subsequent behavior [36–38].
- **Goal Inference.** They distinguish unwilling from unable actors and respond to actions according to the demonstrator's apparent goal and available choices [39, 40].
- **Deception and Manipulation.** Chimpanzees selectively reveal or withhold information and adjust their actions according to what competitors can perceive [9–11].

4.4 Power and Politics

Traditional view.

Power is often inferred from its visible effects, such as dominance, status, authority, influence, and control [73–75], resulting in fragmented accounts rather than a clear understanding of power itself. More broadly, animals, particularly primates, are often intuitively portrayed as brutish and impulsive, overlooking the strategic and interactive nature of their social behavior [14]. Even attacks on neighboring groups reflect sensitivity to potential costs, benefits, and relative advantage [76].

Cross-species view.

Chimpanzees exhibit several constituent features of power and political behavior:

- **Power and Dominance.** Chimpanzee dominance depends not only on physical strength but also on alliances, intimidation, timing, and the management of social relationships [14, 42, 43].
- **Coalition Building.** They form fluid coalitions to support allies, challenge rivals, and reshape dominance hierarchies, often among non-relatives [14–16].
- **Conflict.** Chimpanzees adjust aggression according to status, allies, audience, and risk, and conflicts may be followed by reconciliation or third-party intervention [14, 16, 44].
- **Intergroup Hostility.** Chimpanzee communities conduct coordinated territorial patrols and attacks under numerical advantage, sometimes producing lasting territorial expansion [45, 46, 77, 78].

4.5 Morality and Fairness

Traditional view.

Morality is a strongly positive concept and has accumulated diverse meanings involving right and wrong, virtue, duty, justice, and obligation [79–81]. Its positive status also makes it easy to treat morality as uniquely human, leading to the assumption that animals lack it rather than considering whether more basic forms may exist in other primates [82].

Cross-species view.

Chimpanzees and other primates exhibit several constituent features associated with morality and fairness:

- **Spontaneous Altruism and Helping.** Chimpanzees provide instrumental help in simple, low-cost situations, particularly when recipients clearly signal their needs [48–50].
- **Empathy.** They engage in post-conflict consolation, with bystanders selectively comforting distressed partners [51–53].
- **Fairness and Inequity Aversion.** Primates respond to unequal rewards, while chimpanzees show context-sensitive reactions to relative outcomes and equitable divisions [13, 54, 55].
- **Reconciliation and Relationship Repair.** Former opponents restore affiliative contact after conflicts, while bystanders may console victims and reduce their stress [56–58].

4.6 Shared Biological Substrates

This section examines capacities found in chimpanzees that are also shared across other species:

- **Basic Drives and Preferences.** Chimpanzees form stable preferences from learned associations and select cues linked to previously rewarding outcomes [59].
- **Reproductive Behavior.** Their mating behavior is shaped by dominance, coalition support, mate guarding, social tolerance, and changing group relationships [14].
- **Emotion.** Chimpanzees display emotional responses that support flexible behavioral adjustment, including hesitation, conflict avoidance, and reconciliation [60].
- **Memory.** Chimpanzees and bonobos recognize familiar individuals after separations of more than a decade, demonstrating durable long-term social memory [61].

5 Scope

This framework synthesizes core chimpanzee cognitive capacities into a functionally organized structure for cross-domain and cross-species comparison. It is intended to identify coherent cognitive patterns rather than exhaustively represent every reported behavior or source of empirical variation.

6 Implication: From Assumption-First to Empirical-First Human Science

The broader implication of this framework is methodological. Research on human behavior, cognition, and society has often begun from abstract categories that already impose assumptions about what human capacities are and how they should be interpreted. Cross-species evidence reverses this order by grounding the study of human behavior in observable capacities and their organization, rather than in prior conceptions of what humans are or ought to be.

This empirical-first approach is especially important because the history of primate research has repeatedly challenged assumptions formed before sufficient behavioral evidence was available. Capacities once treated as defining boundaries of humanity—including tool use, fairness, tactical deception, and group hostility—have gradually been identified in nonhuman primates. These findings show that theories of human behavior remain fragile when they begin from what humans are assumed to be, rather than from what humans and related species actually do.

This framework therefore provides an empirical foundation for the study of human behavior. By identifying capacities that extend across species, it provides a basis for defining and investigating human phenomena, while clarifying the foundations from which more complex capacities are built.

7 Future Directions

Future research should begin with cross-species evidence to identify the underlying capacity and the conditions under which it appears, then define the phenomenon precisely and design ecologically grounded tasks that measure it directly rather than relying on vague labels or convenient proxy tasks. For example, culture is defined as group behavior that is transmitted and persists over time. Research can then examine which behaviors are more likely to be transmitted, how transmission occurs, and under what conditions they persist or disappear.

8 Conclusion

Human cognition, behavior, and society are often studied through high-level constructs without first clarifying the roots from which these phenomena arise. As a result, many accounts remain dependent on narrative interpretation and loosely defined concepts, making it difficult to explain how complex human behavior is built from more basic cognitive functions.

This paper proposed a comparative framework for addressing this problem. By aligning core domains of human cognition with functionally comparable capacities observed across species, the framework shows that many capacities often treated as uniquely human can be understood as structured extensions of shared cognitive functions.

More broadly, this comparative framework provides an empirical foundation for moving beyond vague constructs and narrative accounts toward a mechanism-based science of human cognition.

Supplementary information. The Supplementary Information provides detailed descriptions and supporting evidence for the chimpanzee behaviors discussed in this article.

Declarations

Competing interests

The author declares no competing interests.

Ethics approval and consent to participate

Not applicable. This article does not report new research involving human participants or animals.

Data availability

No new data were generated or analyzed in this study.

Author contributions

The author is solely responsible for all aspects of this work.

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